Frictive Policy Optimization for LLMs: Epistemic Intervention, Risk-Sensitive Control, and Reflective Alignment

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Abstract

We propose *Frictive Policy Optimization* (FPO), a framework for learning language model policies that regulate not only what to say, but *when and how to intervene* in order to manage epistemic and normative risk. Unlike standard alignment methods that optimize surface-level preference or task utility, FPO treats clarification, verification, challenge, redirection, and refusal as explicit control actions whose purpose is to shape the evolution of belief, commitment, and uncertainty over time. We formalize alignment as a risk-sensitive epistemic control problem in which intervention decisions are selected based on their expected effect on downstream epistemic quality rather than on immediate reward alone.

We introduce a compact taxonomy of frictive interventions, a structured *friction functional* that operationalizes multiple alignment failure modes, and a unified family of FPO methods spanning reward shaping, preference pairing, group-relative ranking, and risk-conditioned trust regions. We further propose an evaluation framework that measures epistemic competence directly through clarification behavior, calibration, contradiction repair, refusal proportionality, and information efficiency. Together, these results provide a formal and algorithmic foundation for learning agents that are aligned not only in outcome, but in epistemic conduct.

1 Introduction

Large language models (LLMs) are increasingly deployed not merely as generators of text, but as interactive agents expected to collaborate with users over extended dialogues. In such settings, success depends not only on producing fluent or informative responses, but on exercising epistemic judgment: recognizing when a request is underspecified, when assumptions require verification, when a response may be unsafe or normatively problematic, and when contradictions or errors must be repaired over the course of an interaction.

Despite dramatic improvements in instruction following and preference alignment, contemporary alignment paradigms remain poorly equipped to support this form of epistemic competence. Methods such as reinforcement learning from human feedback (RLHF), direct preference optimization (DPO), and group-relative policy optimization (GRPO) primarily optimize for compliance with observed preferences or rankings. They implicitly assume that a response is always required and that the principal challenge is selecting the *best* response among alternatives. As a consequence, these methods systematically conflate epistemic responsibility with surface-level helpfulness.

This limitation manifests in well-documented failure modes. Aligned models frequently respond confidently to underspecified or ambiguous requests, produce answers grounded in unverified assumptions, comply with ill-posed or hazardous instructions, and struggle to repair contradictions across turns. These behaviors are not anomalous; they are structurally induced by training objectives that lack any representation of *intervention* as a legitimate action. From the perspective of such objectives, hesitation, clarification, and refusal appear only as suboptimal answers rather than as rational control decisions.

Recent empirical work reinforces this diagnosis by showing that LLMs, despite their strong generative abilities, systematically struggle with interactional timing – for example, failing to predict appropriate moments to speak in unscripted dialogue (Umair et al., 2024). This highlights a fundamental limitation of alignment methods that optimize only what to say, rather than when and whether to speak at all.

In this paper, we argue that alignment failures of this kind reflect a deeper conceptual gap: the absence of a principled account of *epistemic intervention*. Human collaborators routinely resist immediate task completion in order to improve the epistemic quality of an interaction. They ask clarifying questions, challenge premises, request evidence, redirect ill-posed tasks, or refuse requests that violate safety or normative constraints. Crucially, these behaviors are not expressions of uncertainty alone; they are deliberate acts that regulate commitment, manage risk, and scaffold joint understanding.

We adopt the notion of *friction* not as an obstacle to optimal behavior, but as a constructive control signal that regulates when and how an agent should act under epistemic and normative uncertainty. In human dialogue, productive interaction often depends on well-timed hesitation, clarification, or refusal, rather than immediate task completion. This perspective reframes alignment not as maximizing instantaneous helpfulness, but as managing the dynamics of belief, risk, and coordination over time.

To address this gap, we introduce *Frictive Policy Optimization* (FPO), a general framework for learning policies that strategically deploy epistemic friction. FPO treats intervention types – such as clarification, verification, challenge, redirection, and refusal – as first-class actions. These actions are selected based on their expected impact on epistemic and normative risk, rather than on surface-level preference alone. At the core of FPO is a structured *friction functional* that quantifies epistemic failure modes, including uncertainty miscalibration, contradiction with established context, safety hazards, value conflicts, and expected information gain. This functional serves as a unifying signal across multiple optimization paradigms, enabling friction to be incorporated into reward-based, preference-based, and ranking-based learning methods.

Importantly, FPO is not a single algorithm. It is a family of methods unified by a common theoretical commitment: that resistance to immediate task completion can be rational, learnable, and essential for alignment. This paper builds on a line of work initiated in Pustejovsky and Krishnaswamy (2025), where Frictive Policy Optimization was first introduced as a framework for incorporating epistemic friction into the alignment of large language model agents. That initial work proposed three concrete algorithmic instantiations—Friction-Augmented Rewards (FAR), Friction Preference Pairing (FPP), and Group-Relative Frictive Ranking (GRFR)—and demonstrated their feasibility in agent interaction settings.

The present paper substantially extends this framework along three dimensions. First, we provide a formal treatment of epistemic risk and its surrogate approximation by a structured friction functional. Second, we generalize the original algorithms into a unified control-theoretic framework for learning intervention policies. Third, we introduce a new method, *Friction-Conditioned Trust Regions* (FTR), which regulates policy plasticity itself as a function of epistemic risk. Together, these extensions transform FPO from a heuristic proposal into a principled family of risk-sensitive intervention learning methods.

This paper makes five main contributions. (i) We formalize epistemic intervention in dialogue as a risk-sensitive control problem, in which agents must choose not only what to say, but when to clarify, refuse, defer, or repair. (ii) We define a structured friction functional that serves as a tractable surrogate for latent epistemic risk, decomposing multiple epistemic and normative failure modes into measurable components. (iii) We introduce a unified family of Frictive Policy Optimization methods—FAR, FPP, GRFR, and FTR—that incorporate epistemic friction into learning through rewards, preferences, rankings, and risk-conditioned policy constraints. (iv) We provide formal objectives, optimization procedures, and theoretical analysis showing when and why epistemic interventions such as clarification are optimal under the proposed framework. (v) We propose a trajectory-based evaluation framework that measures clarification competence, repair behavior, proportional refusal, and information efficiency, rather than surface-level helpfulness alone.

Conceptually, this work reframes alignment as a problem of epistemic control rather than static preference satisfaction. Algorithmically, it provides concrete learning methods for inducing selective, risk-calibrated intervention behavior in language models. Our goal is to establish a principled foundation for training and evaluating models that regulate commitment, manage uncertainty, and intervene responsibly in long-horizon interaction.

2 Related Work

This section situates Frictive Policy Optimization (FPO) within four converging lines of research: (i) abstention, deferral, and selective answering in large language models; (ii) epistemic alignment and self-regulation in interactive systems; (iii) decision-theoretic models of clarification and dialogue control; and (iv) risk-sensitive and constrained reinforcement learning. Across these literatures, a common concern is how intelligent agents should regulate commitment under uncertainty—when to answer, when to defer, and when to resist a request altogether. Our framework unifies these themes by treating epistemic intervention as a first-class control problem and by providing a learning-based account of how such interventions can be optimized.

Frictive Policy Optimization (FPO) was first introduced in Pustejovsky and Krishnaswamy (2025), where it was argued that alignment methods based solely on task reward or preference learning neglect a critical epistemic dimension of human–AI collaboration. That paper introduced the core metaphor of *friction* and proposed three initial algorithmic variants: Friction-Augmented Rewards (FAR), Friction Preference Pairing (FPP), and Group-Relative Frictive Ranking (GRFR). Subsequent follow-up work has explored extensions of this idea to epistemic alignment and structured intervention policies (Obiso et al., 2025; Nath and Krishnaswamy, 2025; Nath et al., 2025, 2026). The present paper unifies and generalizes these strands by (i) formalizing latent epistemic risk and its surrogate approximation, (ii) providing a principled decomposition of friction into productive and unproductive components, and (iii) introducing a new class of risk-conditioned regularization methods (FTR) that operate at the level of policy updates rather than learning signals. In contrast to earlier work, which treated friction primarily as an auxiliary reward or preference signal, the framework developed here treats intervention choice as a first-class control problem and situates FPO within the broader literature on risk-sensitive and constrained policy optimization.

Early work on refusal and deferral in dialogue focused on grounding and repair mechanisms in collaborative conversation (Traum, 1994; Clark, 1996). More recent LLM-oriented work treats abstention as a reliability and safety mechanism for selective prediction, rejection, and refusal. A comprehensive survey by Wen et al. (2025) categorizes abstention methods by lifecycle stage (pretraining, alignment, inference) and proposes standardized evaluation frameworks. We differ from this literature in two respects. First, we treat abstention and refusal not as special-case safety filters but as first-class control actions within a unified policy. Second, we optimize abstention jointly with clarification and self-correction under a single risk-sensitive objective, rather than introducing abstention only as an inference-time heuristic. Recent work on input clarification and uncertainty decomposition (Hou et al., 2024) further supports the view that clarification is an active mechanism for managing epistemic uncertainty rather than a post-hoc repair.

A growing body of work frames alignment in explicitly epistemic terms. Clark et al. (2025) propose an Epistemic Alignment Framework that identifies challenges such as ambiguity resolution, abstention, uncertainty expression, and self-correction as central to aligned behavior. Constitutional AI (Bai et al., 2022) operationalizes self-regulation through post-generation critique and revision guided by normative principles. Our framework is complementary but differs in formalization. Whereas Constitutional AI treats critique and revision as post-hoc correction mechanisms, we treat clarification, refusal, and self-correction as actions chosen directly by a control policy. Likewise, while epistemic alignment frameworks provide conceptual taxonomies of challenges, we provide a unified decision-theoretic objective and concrete optimization methods for learning epistemic interventions.

Clarification has long been studied in probabilistic dialogue systems and decision-theoretic models. In POMDP-based dialogue management, belief states summarize interaction histories and policies are optimized over these latent states (Young et al., 2013; Williams and Young, 2007; Traum, 1994). Decision-theoretic models of question generation explicitly optimize expected information gain or entropy reduction to decide when to ask clarifying questions (Gervits et al., 2021). Recent work such as SAGE-Agent formulates clarification as a POMDP with expected value of information objectives and trains policies that reward clarification under uncertainty (Suri et al., 2025). Our formulation generalizes this line of work by embedding clarification within a broader friction functional that also captures contradiction, hazard, and value conflict, and by optimizing clarification jointly with refusal and self-correction within a single policy.

Our objective also builds on a large literature on risk-sensitive and robust reinforcement learning. Classical work on variance-aware and CVaR-based objectives develops criteria for optimizing tail risk and worst-case performance (Tamar et al., 2015; Chow et al., 2015). More recent work provides general reductions from optimized certainty equivalents to standard RL algorithms (Wang et al., 2024). We differ from this literature in the nature